Karl El Hajal

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RESEARCH INTERESTS

Speech & Audio AI, Representation Learning, Self-Supervised Learning, Text-to-Speech, Voice Conversion, Pathological Speech, Spoken Language Models, Music Generation.

EDUCATION

PhD in Electrical Engineering — Mar. 2024–2028 (expected)
EPFL, Swiss Federal Institute of Technology Lausanne — Lausanne, Switzerland
Supervisor: Dr. Mathew Magimai-Doss. Thesis: "Pathological Speech Synthesis".

MSc in Electrical and Electronic Engineering — Sep. 2020–Sep. 2022
EPFL, Swiss Federal Institute of Technology Lausanne — Lausanne, Switzerland

BE in Computer and Communications Engineering — Sep. 2015–Jun. 2019
American University of Beirut — Beirut, Lebanon
Visiting student at UC Berkeley (2017): Software Engineering.

EXPERIENCE

Idiap Research Institute — Martigny, Switzerland
PhD Researcher — Mar. 2024–Present

- Developed and published low-resource and efficient methods for zero-shot TTS and unsupervised voice conversion; improved ASR for speakers with more severe dysarthria.
- Trained novel speech representation models for joint speech analysis and synthesis.
- Researching pathological speech synthesis to address scarce clinical data.

Research Assistant — Mar. 2023–Mar. 2024

- Worked on a production multilingual, multi-speaker emotional TTS system; benchmarked models and studied code-switching and context-dependent text normalization for French, German, and Italian.

Logitech — Lausanne, Switzerland
Audio Machine Learning Research Intern — Aug. 2021–Aug. 2022

- Developed MOSRA for joint speech-quality and room-acoustics assessment; combined it with BYOL-S to cut model size by 40–60x and latency by 30x versus a leading baseline.
- Built PyTorch pipelines for multimodal affective computing across speech, video, and language.

Murex — Beirut, Lebanon
Full-time Software Engineer, C/C++ — Jun. 2019–Sep. 2020

- Developed and tested a parallel real-time graph engine for financial models, modernized legacy C code, expanded test coverage, and served as Scrum Master.

University of Illinois Urbana-Champaign — Urbana, IL, USA
Visiting Research Intern — Jun. 2018–Sep. 2018

- Researched the effectiveness of runtime monitoring for formal verification using JavaMOP.

American University of Beirut — Beirut, Lebanon
Research Assistant — Jan. 2018–Jun. 2019

- Developed SOQAL for open-domain Arabic question answering.
- Crowdsourced a large-scale Arabic reading-comprehension dataset.

AWARDS & ACHIEVEMENTS

EPFL School of Engineering Teaching Assistant Award (2026)

Idiap Create Challenge Winner (2025)
Co-created SYNTHIA, an AI music startup for live, responsive jam partners.

ICASSP Top 3% Paper Recognition (2023)

AUB Most Innovative Software Project (2019)
Final-year project on open-domain Arabic question answering using large language models.

SELECTED PUBLICATIONS

Concatenative Neural Speech Synthesis
Karl El Hajal, Sanat Kumar Agrawal, Enno Hermann, and Mathew Magimai-Doss.
IEEE Spoken Language Technology Workshop (SLT), 2026.

Unsupervised Rhythm and Voice Conversion to Improve ASR on Dysarthric Speech
Karl El Hajal, Enno Hermann, Sevada Hovsepyan, and Mathew Magimai-Doss.
Interspeech, 2025.

kNN Retrieval for Simple and Effective Zero-Shot Multi-speaker Text-to-Speech
Karl El Hajal, Ajinkya Kulkarni, Enno Hermann, and Mathew Magimai-Doss.
NAACL, 2025.

Fine-Tuning Pretrained Models with NVIB for Improved Generalisation
Fabio Fehr, Alina Elena Baia, Xiaoguang Chang, Andrei C. Coman, Karl El Hajal, Dina El Zein, Shashi Kumar, Juan Zuluaga-Gomez, Andrea Cavallaro, Damien Teney, and James Henderson.
SCSL Workshop at ICLR, 2025.

Idiap kNN-TTS System for the Blizzard Challenge 2025
Enno Hermann, Karl El Hajal, Ajinkya Kulkarni, and Mathew Magimai-Doss.
Blizzard Challenge Workshop, 2025.

Efficient Speech Quality Assessment Using Self-Supervised Framewise Embeddings
Karl El Hajal, Zihan Wu, Neil Scheidwasser-Clow, Gasser Elbanna, and Milos Cernak.
ICASSP, 2023 – Top 3% Paper Recognition.

Personalized Task Load Prediction in Speech Communication
Robert P. Spang, Karl El Hajal, Sebastian Moller, and Milos Cernak.
ICASSP, 2023.

BYOL-S: Learning Self-Supervised Speech Representations by Bootstrapping
Gasser Elbanna, Neil Scheidwasser-Clow, Mikolaj Kegler, Pierre Beckmann, Karl El Hajal, and Milos Cernak.
Proceedings of Machine Learning Research, vol. 166, 2022.

MOSRA: Joint Mean Opinion Score and Room Acoustics Speech Quality Assessment
Karl El Hajal, Milos Cernak, and Pablo Mainar.
Interspeech, 2022.

Neural Arabic Question Answering
Hussein Mozannar, Karl El Hajal, Elie Maamary, and Hazem Hajj.
WANLP at ACL, 2019.

PREPRINTS

OLIVE: View-Augmented Latent Prediction with Waveform Reconstruction for Speech SSL
Karl El Hajal and Mathew Magimai-Doss.
2026. Under review.

ClusterVAD: Threshold-Free Unsupervised Voice Activity Detection with Pre-Trained Neural Representations
Karl El Hajal and Mathew Magimai-Doss.
2026. Under review.

TEACHING

EPFL – EE-554 Automatic Speech Processing
Teaching Assistant 2023–Present

- Delivered lectures on neural methods for speech synthesis and self-supervised speech representation learning; prepared laboratory exercises, maintained course materials, and graded examinations.
- Co-supervised a Master's semester project that led to an accepted conference paper.

Unidistance – Speech and Audio Processing
Teaching Assistant 2023–2024

TECHNICAL SKILLS

Programming	Python, C/C++, Java, C#, MATLAB, TypeScript, JavaScript.
Machine learning	PyTorch, TensorFlow, Hugging Face, Weights & Biases.
Systems and tools	Unix/Linux, Git, Docker, Google Cloud, AWS, LaTeX.
Languages	English, French, and Arabic.

REVIEWING

ICASSP, ACL Rolling Review, *Speech Communication*, *Computer Speech & Language* (2026).